



Project Proposal

Team Members & Project Title

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[Insert Project Title Here] Mobile Payment Fraud Detection Using Machine Learning					

Note: Team member names are listed alphabetically by last name and do not reflect contribution level.

Topic & Summary

Topic of Interest

[Write 1-2 paragraphs here. Briefly describe the topic your team is investigating.]

Our team is investigating how machine learning can be used to detect fraudulent mobile payment transactions. Mobile payment platforms process large numbers of transactions, so it can be difficult to identify suspicious behavior manually or through simple rule-based systems. This project will use the PaySim dataset, a synthetic mobile-money transaction dataset, to explore whether transaction features such as transaction type, amount, sender/receiver balances, and fraud labels can help classify transactions as fraudulent or non-fraudulent.

Because real financial transaction data is private and difficult to access, synthetic datasets like PaySim provide a safer way to study fraud detection while still discussing important limitations. Our goal is not to build a production-ready fraud system, but to compare several supervised machine-learning models and evaluate which approach performs best in a simulated setting.

Summary: 1. Why Is This Topic Important? 2. How Will Data Be Used? 3. Potential Impact

1. [Write 1 paragraph here. Explain why this topic matters to individuals, communities, industries, or society.]

Fraud detection is important because fraudulent payments can cause financial loss, reduce trust in digital payment systems, and create security risks for users and financial institutions. Since fraud is usually rare compared with normal transactions, a model must do more than simply achieve high accuracy. A useful fraud detection system should identify suspicious transactions while also minimizing false alerts that may inconvenience legitimate users.

2. [Write 1 paragraph here. Describe what data will be analyzed and how it will help answer your research question.]

We will analyze the PaySim dataset, which contains simulated mobile payment transaction records. The dataset includes features such as transaction type, transaction amount, account balances before and after the transaction, and whether the transaction is labeled as fraud. We will clean the data, split it into training and testing sets, train classification models, and evaluate the models using precision, recall, F1-score, confusion matrix, and possibly PR-AUC instead of accuracy alone.

3. [Write 1 paragraph here. Describe how your findings could help people make decisions, improve understanding, identify trends, or solve real-world problems.]

Our findings could help show which machine-learning models are more effective for identifying risky mobile payment



patterns in a simulated environment. This project may help financial platforms think about how to reduce fraud losses while also avoiding unnecessary false alerts for normal users. It can also help our team understand the ethical limits of automated fraud detection, especially when the dataset is synthetic and may not fully represent real-world users.

Machine Learning Algorithm(s)

Logistic Regression

We will use logistic regression as a baseline model because it is simple, interpretable, and commonly used for binary classification problems.

Random Forest

We will use random forest because it can capture non-linear patterns and feature interactions in transaction data better than a single linear model.

Gradient Boosting / XGBoost

We will explore gradient boosting or XGBoost because boosted tree models often perform well on structured/tabular fraud detection data. If XGBoost is too advanced for our timeline, we may use a decision tree as a simpler alternative.

Research Question

Possible question: Can we classify payments/transactions as fraudulent or not fraudulent?

Possible question: Can we predict/detect fraudulent payment based on ...?

Possible question: Which machine-learning model can best identify fraudulent mobile payment transactions while minimizing false alerts?

Which supervised machine-learning model can best detect fraudulent versus non-fraudulent mobile payment transactions in the PaySim dataset, where “best” is defined as achieving high fraud-class recall while maintaining reasonable precision and a manageable false positive rate?

Dataset(s)

- **Name:** PaySim
- **Source:** Kaggle; created using the PaySim mobile-money simulator
- **Link:**
<https://www.kaggle.com/datasets/ealaxi/paysim1>

- **Name:** IEEE-CIS Fraud Detection
- **Source:** Kaggle; data comes from Vesta's real-world e-commerce transactions
- **Link:**
<https://www.kaggle.com/competitions/ieee-fraud-detection>

Training & Testing Considerations

The PaySim dataset contains over six million simulated mobile payment transactions, providing enough data for training and testing. Because fraudulent transactions are rare, we will use a stratified train-test split, such as 80% training and 20% testing, to maintain the fraud ratio across both sets. Given the dataset's massive size, training advanced models on local machines with standard 8GB memory limits will cause system performance issues. Therefore, we will initially extract a smaller, representative subset for local script testing, exploratory data analysis, and feature engineering before running the final models.

To help the models detect suspicious patterns, we plan to engineer new features to capture anomalies. For example, we will calculate balance errors to verify if the transaction amount matches the sender's actual balance change. We will also



flag transactions that attempt to transfer the exact total of the account balance, which is a common cash-out fraud indicator. We will evaluate models using precision, recall, F1-score, confusion matrix, and PR-AUC. If needed, we may use class weights or resampling methods on the training data only. We will avoid using customer or account ID fields directly as predictive features to prevent the model from memorizing transaction identities instead of learning general fraud patterns.

Because fraud detection is an imbalanced classification problem, we will not select the final model based on accuracy alone. Instead, we will prioritize recall for the fraud class, because missing actual fraud cases can be costly. At the same time, we will monitor precision and the false positive rate to avoid creating too many unnecessary alerts for normal users. If two models have similar fraud recall, we will prefer the model with the stronger F1-score, PR-AUC / average precision, and more interpretable results. We will compare models using recall, precision, F1-score, confusion matrix, false positive rate, and PR-AUC / average precision rather than accuracy alone.

Sources of Bias

[Bias #1. Identify possible sources of bias in your project.]

Synthetic Data Bias

PaySim is a simulated dataset, so it may not fully capture all real-world fraud behaviors, user patterns, or banking system differences. Moreover, there may be an induced sterility, paucity of noise dynamicity, or bias with data distribution.

[Bias #2. **Examples:** Underrepresentation of certain groups, Missing or incomplete data, Historical bias in data collection, Geographic bias, Sampling bias, Labeling bias, Measurement bias]

Geographic / Context Bias

The PaySim simulator is based on mobile money transaction patterns from a specific context, so the results may not generalize to all countries, banks, or payment platforms.

[Bias #3]

Class Imbalance / Label Bias

Fraudulent transactions are much less common than normal transactions, so models may become biased toward predicting the majority class. A model may appear highly accurate by predicting most transactions as non-fraudulent, even if it misses many actual fraud cases. Because the labels come from a simulation process, they may also not reflect every type of real-world fraud.

Mitigation Strategies

[Mitigation Strategy #1. Describe at least two practical ways your team will reduce the negative impact of bias.]

We will clearly state that PaySim is a synthetic dataset, so our results should be interpreted as exploratory rather than as a real-world fraud detection system ready for deployment. We will avoid making claims about specific populations, real customers, or real financial institutions.

[Mitigation Strategy #2]

We will use stratified splitting to maintain the fraud ratio across training and testing sets. We will evaluate models with precision, recall, F1-score, confusion matrix, and possibly PR-AUC instead of accuracy alone. If needed, we may also use class weights or resampling methods on the training data to reduce the effect of class imbalance.

Citations

[Citation. Include at least 3-4 scholarly, reputable, or news sources related to your topic.]	López-Rojas, E. A., Elmir, A., & Axelsson, S. (2016). PaySim: A financial mobile money simulator for fraud detection. https://www.msc-les.org/proceedings/ems/2016/EMSS2016_249.pdf	Kaggle. Synthetic Financial Datasets For Fraud Detection. https://www.kaggle.com/datasets/ealaxi/paysim1	Isangediok, M., & Gajamannage, K. (2022). Fraud Detection Using Optimized Machine Learning Tools Under Imbalance Classes. arXiv. https://arxiv.org/abs/2209.01642	Google Developers. Classification: Accuracy, recall, precision, and related metrics. https://developers.google.com/machine-learning/crash-course/classification/accuracy-precision-recall	Scikit-learn. Precision-Recall. Scikit-learn documentation. https://scikit-learn.org/stable/auto_examples/model_selection/plot_precision_recall.html	[Citation]
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